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TREND-BASED MOCK PAPER

GATE Computer Science and Information Technology

Mock Paper — 04

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Computer Science and Information Technology
Paper:	Mock 04
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [2-MCQ] [2 marks]

In a class of 60 students, 30 play cricket, 25 play football, and 15 play both. How many play neither?

- (A) 20
- (B) 25
- (C) 30
- (D) 35

Q2. [2-NAT] [2 marks]

Antonym of 'ephemeral':

(Numerical Answer Type — Enter the numerical value)

Section B: Computer Science and Information Technology

Q3. [3-MCQ] [4 marks]

Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2,3,4?

- (A) FIFO
- (B) LRU
- (C) Optimal
- (D) None of these

Q4. [2-MSQ] [2 marks]

Semaphore with value 1 is:

- (A) Counting
- (B) Binary (mutex)
- (C) Monitor
- (D) Lock variable

Q5. [2-MCQ] [2 marks]

Banker's algorithm is for:

- (A) Detection
- (B) Avoidance
- (C) Prevention
- (D) Recovery

Q6. [2-NAT] [2 marks]

Round-robin uses:

(Numerical Answer Type — Enter the numerical value)

Q7. [2-NAT] [2 marks]

Inode contains:

(Numerical Answer Type — Enter the numerical value)

Q8. [2-MCQ] [2 marks]

TLB stands for:

- (A) Translation Lookaside Buffer
- (B) Time Local Buffer
- (C) Total Logic Block
- (D) None

Q9. [2-MSQ] [2 marks]

Which of the following is NOT a necessary condition for deadlock?

- (A) Mutual Exclusion
- (B) Hold and Wait
- (C) Preemption
- (D) Circular Wait

Q10. [1-MCQ] [1 mark]

Critical section problem solution requirements:

- (A) Mutual exclusion only
- (B) Progress and bounded waiting also
- (C) No requirements
- (D) Only progress

Q11. [1-MCQ] [1 mark]

Demand paging requires:

- (A) No MMU
- (B) Hardware support (MMU)
- (C) No secondary storage
- (D) Swap space not needed

Q12. [2-NAT] [2 marks]

Protocol to find MAC from IP:

(Numerical Answer Type — Enter the numerical value)

Q13. [2-MCQ] [2 marks]

Which protocol operates at the application layer of the TCP/IP model?

- (A) TCP

- (B) UDP
- (C) HTTP
- (D) IP

Q14. [2-MCQ] [2 marks]

CSMA/CD is used in:

- (A) Token Ring
- (B) Ethernet
- (C) WiFi
- (D) Bluetooth

Q15. [2-NAT] [2 marks]

Asymmetric encryption:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-NAT] [2 marks]

OSPF is:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

TCP connection establishment uses:

- (A) 2-way
- (B) 3-way handshake
- (C) 4-way
- (D) 1-way

Q18. [1-MCQ] [1 mark]

In Go-Back-N, sender window size = 7. Sequence numbers needed:

- (A) 7
- (B) 8
- (C) 14
- (D) 6

Q19. [1-NAT] [1 mark]

CSMA/CD after 3rd collision, prob of choosing slot 5 ($k=5$):

(Numerical Answer Type — Enter the numerical value)

Q20. [1-MCQ] [1 mark]

Which uses Bellman-Ford?

- (A) OSPF
- (B) RIP
- (C) BGP
- (D) IS-IS

Q21. [1-MCQ] [1 mark]

Stop-and-Wait utilization with $RTT=100ms$, $t_{tx}=1ms$:

- (A) 1%
- (B) 1/101
- (C) 99%
- (D) 100%

Q22. [1-MCQ] [1 mark]

Maximum data rate (Nyquist) for 3kHz bandwidth:

- (A) 6000 bps
- (B) 3000 bps
- (C) 12000 bps
- (D) 600 bps

Q23. [1-NAT] [1 mark]

CIDR block 200.10.0.0/20 contains how many class C addresses?

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

Kruskal uses:

- (A) DFS
- (B) BFS
- (C) Union-Find
- (D) Dijkstra

Q25. [2-MCQ] [2 marks]

Merge sort space:

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n^2)$

Q26. [2-MCQ] [2 marks]

Time complexity of binary search:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(n \log n)$
- (D) $O(1)$

Q27. [2-NAT] [2 marks]

BFS uses:

(Numerical Answer Type — Enter the numerical value)

Q28. [3-MCQ] [4 marks]

The time complexity of Dijkstra's algorithm using a binary heap is:

- (A) $O(V^2)$
- (B) $O(E \log V)$
- (C) $O(V \log V + E)$
- (D) $O(V \log V)$

Q29. [2-MCQ] [2 marks]

LCS uses:

- (A) Greedy
- (B) Divide and Conquer
- (C) Dynamic Programming
- (D) Backtracking

Q30. [1-MCQ] [1 mark]

Which sort is not comparison-based?

- (A) Merge sort
- (B) Heap sort
- (C) Counting sort
- (D) Quick sort

Q31. [1-NAT] [1 mark]

Minimum spanning tree with Kruskal's on sorted edges:

(Numerical Answer Type — Enter the numerical value)

Q32. [1-MCQ] [1 mark]

Floyd-Warshall computes all-pairs shortest paths in:

- (A) $O(n^3)$
- (B) $O(n^2 \log n)$
- (C) $O(nm)$
- (D) $O(n^2)$

Q33. [2-MCQ] [2 marks]

Determinant of 2×2 identity:

- (A) 0
- (B) 1
- (C) 2
- (D) -1

Q34. [2-MCQ] [2 marks]

Probability of at least one head in 3 coin tosses:

- (A) $1/8$
- (B) $3/8$
- (C) $7/8$
- (D) $1/2$

Q35. [2-MCQ] [2 marks]

Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

- (A) 6
- (B) 7
- (C) 14
- (D) 15

Q36. [2-NAT] [2 marks]

Derivative of x^3 at $x=2$:

(Numerical Answer Type — Enter the numerical value)

Q37. [2-MCQ] [2 marks]

Mean of binomial $n=10, p=0.5$:

- (A) 2
- (B) 5
- (C) 10
- (D) 0.5

Q38. [1-MCQ] [1 mark]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q39. [1-MCQ] [1 mark]

The Laplace transform of $t \cdot e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q40. [2-MCQ] [2 marks]

Infix to postfix uses:

- (A) Queue
- (B) Stack
- (C) Tree
- (D) Graph

Q41. [2-NAT] [2 marks]

What is the time complexity of inserting at the beginning of a linked list?

(Numerical Answer Type — Enter the numerical value)

Q42. [2-NAT] [2 marks]

Inorder traversal of BST gives:

(Numerical Answer Type — Enter the numerical value)

Q43. [2-MCQ] [2 marks]

Avg hash lookup:

- (A) $O(n)$
- (B) $O(\log n)$
- (C) $O(1)$
- (D) $O(n^2)$

Q44. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q45. [1-MCQ] [1 mark]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \cdot 2$
- (D) $P \cdot r/100$

Q46. [1-NAT] [1 mark]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q47. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
 - (B) $2n$
 - (C) $n(n+1)$
 - (D) $n(n-1)$
-

Q48. [1-MCQ] [1 mark]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q49. [1-MCQ] [1 mark]

Compound interest: P at $r\%$ for 2 years =

- (A) $P(1+r/100)^2$
 - (B) $P(1+2r/100)$
 - (C) $P + Pr/100 \times 2$
 - (D) $P \times r/100$
-

Q50. [1-MCQ] [1 mark]

Interrupt-driven I/O vs DMA:

- (A) DMA is faster for single byte
 - (B) DMA is better for block transfer
 - (C) Interrupt is better for large data
 - (D) Both same
-

Q51. [1-NAT] [1 mark]

Mod-12 counter requires minimum flip-flops:

(Numerical Answer Type — Enter the numerical value)

Q52. [1-NAT] [1 mark]

Cache 64 blocks, 4-way set assoc, block 16B. Address 200 maps to:

(Numerical Answer Type — Enter the numerical value)

Q53. [1-MCQ] [1 mark]

32KB direct-mapped cache, 64B blocks. Tag bits for 32-bit address:

- (A) 19
 - (B) 20
 - (C) 17
 - (D) 22
-

Q54. [1-MCQ] [1 mark]

2K x 8 ROM has how many address lines?

- (A) 2

- (B) 8
- (C) 11
- (D) 2048

Q55. [2-MSQ] [2 marks]

Flip-flop stores:

- (A) 1 bit
- (B) 2 bits
- (C) 4 bits
- (D) 8 bits

Q56. [1-MCQ] [1 mark]

The number of Boolean functions of n variables is:

- (A) 2^n
- (B) $2^{(2^n)}$
- (C) $2^{(n^2)}$
- (D) n^2

Q57. [1-NAT] [1 mark]

Full adder has how many inputs:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-MCQ] [1 mark]

B+ tree leaf nodes are linked for:

- (A) Faster insert
- (B) Range queries
- (C) Faster delete
- (D) Balance

Q59. [1-NAT] [1 mark]

R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$. Candidate keys:

(Numerical Answer Type — Enter the numerical value)

Q60. [1-NAT] [1 mark]

Which concurrency protocol avoids cascading aborts?

(Numerical Answer Type — Enter the numerical value)

Q61. [1-NAT] [1 mark]

B+ tree leaf nodes are linked for:

(Numerical Answer Type — Enter the numerical value)

Q62. [1-MCQ] [1 mark]

Halting problem:

- (A) Decidable
- (B) Undecidable
- (C) Semi-decidable
- (D) None

Q63. [1-NAT] [1 mark]

PDA recognizes:

(Numerical Answer Type — Enter the numerical value)

Q64. [1-MCQ] [1 mark]

Which language is regular?

- (A) Equal 0s and 1s
- (B) Strings ending in ab
- (C) Palindromes
- (D) $a^n b^n$

Q65. [1-MCQ] [1 mark]

Lexical analyzer outputs:

- (A) Tokens
 - (B) Parse tree
 - (C) Intermediate code
 - (D) Assembly
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	2MCQ	A	2
2	2NAT	B	2
3	2MCQ	B	2
4	2MCQ	C	2
5	2MCQ	C	2
6	2NAT	B	2
7	2MCQ	B	2
8	2MCQ	B	2
9	2NAT	B	2
10	2NAT	A	2
11	2MCQ	C	2
12	2MCQ	C	2
13	2MCQ	C	2

14	2MCQ	B	2
15	2NAT	B	2
16	3MCQ	C	4
17	2MCQ	C	2
18	3MCQ	C	4
19	2MSQ	B	2
20	2MCQ	B	2
21	2NAT	A	2
22	2NAT	B	2
23	2MCQ	A	2
24	2MSQ	C	2
25	2NAT	B	2
26	2MCQ	C	2
27	2MCQ	B	2
28	2NAT	C	2

29	2NAT	B	2
30	2MCQ	B	2
31	2MSQ	A	2
32	1MCQ	B	1
33	1NAT	B	1
34	1MCQ	B	1
35	1NAT	B	1
36	1MCQ	B	1
37	1MCQ	A	1
38	1MCQ	A	1
39	1MCQ	B	1
40	1MCQ	B	1
41	1MCQ	A	1
42	1MCQ	B	1
43	1MCQ	B	1

44	1NAT	B	1
45	1MCQ	B	1
46	1NAT	B	1
47	1NAT	C	1
48	1MCQ	B	1
49	1MCQ	C	1
50	1MCQ	B	1
51	1MCQ	A	1
52	1NAT	B	1
53	1NAT	A	1
54	1NAT	B	1
55	1NAT	A	1
56	1MCQ	A	1
57	1NAT	A	1
58	1MCQ	A	1

59	1MCQ	A	1
60	1MCQ	B	1
61	1MCQ	C	1
62	1NAT	B	1
63	1MCQ	A	1
64	1MCQ	A	1
65	1MCQ	A	1

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Probability | Difficulty: moderate

Q5.

Answer: C

Solution:

Correct Answer: C

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Calculus | Difficulty: moderate

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Probability | Difficulty: difficult

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Stacks | Difficulty: easy

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Programming and Data Structures | Topic: Linked List | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Programming and Data Structures | Topic: Trees | Difficulty: moderate

Q11.

Answer: C

Solution:

Correct Answer: C

Subject: Programming and Data Structures | Topic: Hashing | Difficulty: easy

Q12.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Greedy | Difficulty: moderate

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Divide and Conquer | Difficulty: difficult

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Complexity Analysis | Difficulty: difficult

Q15.

Answer: B

Solution:

Correct Answer: B

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: moderate

Q16.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Graph Algorithms | Difficulty: easy

Q17.

Answer: C

Solution:

Correct Answer: C

Subject: Algorithms | Topic: Dynamic Programming | Difficulty: difficult

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Memory Management | Difficulty: moderate

Q19.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Synchronization | Difficulty: moderate

Q20.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: Deadlocks | Difficulty: easy

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Process Scheduling | Difficulty: difficult

Q22.

Answer: B

Solution:

Correct Answer: B

Subject: Operating Systems | Topic: File Systems | Difficulty: moderate

Q23.

Answer: A

Solution:

Correct Answer: A

Subject: Operating Systems | Topic: Virtual Memory | Difficulty: moderate

Q24.

Answer: C

Solution:

Correct Answer: C

Subject: Operating Systems | Topic: Deadlocks | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Network Layer | Difficulty: easy

Q26.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Application Layer | Difficulty: moderate

Q27.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Data Link Layer | Difficulty: easy

Q28.

Answer: C

Solution:

Correct Answer: C

Subject: Computer Networks | Topic: Security | Difficulty: moderate

Q29.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Routing | Difficulty: moderate

Q30.

Answer: B

Solution:

Correct Answer: B

Subject: Computer Networks | Topic: Transport Layer | Difficulty: difficult

Q31.

Answer: A

Solution:

Correct Answer: A

Subject: Digital Logic | Topic: Sequential Circuits | Difficulty: moderate

Q32.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Boolean Algebra | Difficulty: easy

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Logic | Topic: Combinational Circuits | Difficulty: difficult

Q34.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Decidability | Difficulty: difficult

Q35.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Context Free Languages | Difficulty: moderate

Q36.

Answer: B

Solution:

Correct Answer: B

Subject: Theory of Computation | Topic: Regular Languages | Difficulty: moderate

Q37.

Answer: A

Solution:

Correct Answer: A

Subject: Compiler Design | Topic: Lexical Analysis | Difficulty: moderate

Q38.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q39.

Answer: B

Solution:

DMA: direct memory access. Better for block transfers (no CPU per byte). Interrupt per byte.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: difficult

Q40.

Answer: B

Solution:

CS requirements: (1) Mutual exclusion, (2) Progress (no deadlock), (3) Bounded waiting (no starvation).

Subject: Operating Systems | Topic: Operating Systems | Difficulty: easy

Q41.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q42.

Answer: B

Solution:

Window = N: need N+1 sequence numbers to distinguish. For N=7: need 8 (3 bits).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: easy

Q43.

Answer: B

Solution:

B+ tree: all data at leaves, linked sequentially. Enables efficient range scans without root-to-leaf traversal.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q44.

Answer: B

Solution:

Closures: (AB)+=ABCD, (BD)+=ABCD, (BC)+=ABCD. Minimal: AB and BD.

Subject: DBMS | Topic: DBMS | Difficulty: moderate

Q45.

Answer: B

Solution:

$P(A|B) = P(A \cap B) / P(B) = 0.2 / 0.4 = 0.5$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

Need $2^n \geq 12$. $n=4$ gives 16 states. Minimum 4 flip-flops.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q47.

Answer: C

Solution:

After 3 collisions, $k=3$. Random from 0..7. $P(\text{slot } 5) = 1/8$.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q48.

Answer: B

Solution:

RIP: distance vector, Bellman-Ford. OSPF: link-state (Dijkstra).

Subject: Computer Networks | Topic: Computer Networks | Difficulty: easy

Q49.

Answer: C

Solution:

Counting sort: $O(n+k)$, uses array indexing not comparisons. Others are comparison-based: $O(n \log n)$ lower bound.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q50.

Answer: B

Solution:

Utilization = $t_{tx} / (t_{tx} + RTT) = 1/101$.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q51.

Answer: A

Solution:

Nyquist: max data rate = $2 \cdot B \cdot \log_2(L)$. For binary ($L=2$): $2 \cdot 3000 \cdot 1 = 6000$ bps.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: easy

Q52.

Answer: B

Solution:

Strict 2PL: holds all locks until commit. Prevents dirty reads, cascading aborts, and unrepeatable reads.

Subject: DBMS | Topic: DBMS | Difficulty: easy

Q53.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18$.

Subject: Data Structures | Topic: Data Structures | Difficulty: easy

Q54.

Answer: B

Solution:

If edge weights are equal, multiple MSTs possible. Kruskal picks arbitrarily among equal-weight edges.

Subject: Algorithms | Topic: Algorithms | Difficulty: moderate

Q55.

Answer: A

Solution:

Block addr = $200/16 = 12$. Sets = $64/4 = 16$. Set = $12 \bmod 16 = 12$. Offset = $200 \bmod 16 = 8$.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: difficult

Q56.

Answer: A

Solution:

Floyd-Warshall: $O(n^3)$ DP over all intermediate vertices.

Subject: Algorithms | Topic: Algorithms | Difficulty: easy

Q57.

Answer: A

Solution:

/20 spans $2^{(24-20)} = 16$ class C (/24) networks.

Subject: Computer Networks | Topic: Computer Networks | Difficulty: moderate

Q58.

Answer: A

Solution:

Offset = 6, index = $\log_2(32768/64) = 9$. Tag = $32-9-6 = 17$. Hmm, 17 is option C. Wait let me recalculate.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: moderate

Q59.

Answer: A

Solution:

$1+3+5+\dots+(2n-1) = n^2$.

Subject: Data Structures | Topic: Data Structures | Difficulty: difficult

Q60.

Answer: B

Solution:

Demand paging: MMU provides valid-invalid bits. Page fault on invalid access.

Subject: Operating Systems | Topic: Operating Systems | Difficulty: easy

Q61.

Answer: C

Solution:

2K = 2048 locations. Need $\log_2 = 11$ address lines.

Subject: Computer Organization | Topic: Computer Organization | Difficulty: difficult

Q62.

Answer: B

Solution:

B+ tree: all data at leaves, linked sequentially. Enables efficient range scans without root-to-leaf traversal.

Subject: DBMS | Topic: DBMS | Difficulty: difficult

Q63.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q64.

Answer: A

Solution:

CI = $P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Data Structures | Topic: Data Structures | Difficulty: moderate

Q65.

Answer: A

Solution:

$L\{f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Operating Systems: 9 questions, 18 marks
Computer Networks: 12 questions, 18 marks
Algorithms: 9 questions, 17 marks
Engineering Mathematics: 7 questions, 12 marks
Programming and Data Structures: 4 questions, 8 marks
Data Structures: 6 questions, 6 marks
Computer Organization: 5 questions, 5 marks
General Aptitude: 2 questions, 4 marks
Digital Logic: 3 questions, 4 marks
DBMS: 4 questions, 4 marks
Theory of Computation: 3 questions, 3 marks
Compiler Design: 1 questions, 1 marks

Difficulty Distribution:

Easy: 16 | Moderate: 36 | Difficult: 13

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.